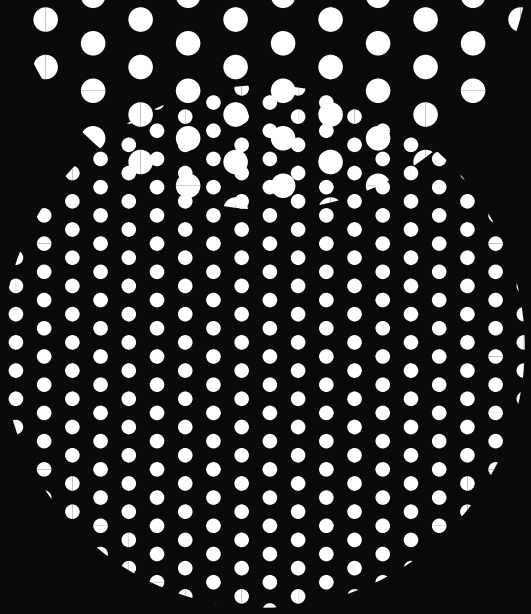


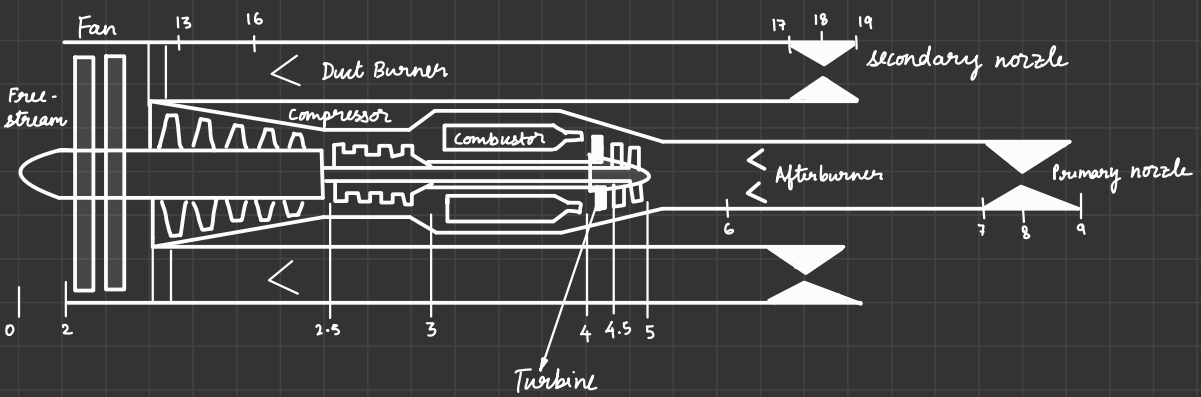


Chp 8 Engine Performance Analysis

PCA ⟶ Rubber Engine {Not sized}
Fuel flow rate ⟶ Main variable {Directly affects T_{t4}}
↳ Throttle setting

Interaction & performance of compressor - burner - Turbine combo
↓
Pumping characteristics

Principal effects of engine performance occur largely due to changes in propulsive efficiency & thermal efficiency.



station numbering for 2-spool Gas turbine engine

8.1.1 Reference values & EPA assumptions

$f(\tau, \pi) = \text{const.}$
represents a relationship b/w the two engine variables τ & π at a steady-state operating pt.

$$f(\tau, \pi) = f(\tau_R, \pi_R) = \text{const.}$$

since $f(\tau, \pi)$ applies to the engine at all operating pts.
sea level static (SLS) is the normal reference cond^n (disign pt.) for the value of gas turbine variables.

$T_{ref} = 288.2\ K$, $P_{ref} = 101,300\ Pa$

Assumptions: 1) Flow is choked at HPT entrance nozzle, LPT entrance nozzle, & primary exit nozzle. Bypass duct also choked.

2) π_b, π_n, π_{fn} do not change from their ref. values.

3) Component efficiencies do not change from their ref. values ($\eta_{cL}, \eta_{cH}, \eta_f, \eta_b, \eta_{tH}, \eta_{tL}, \eta_{mH}$ & η_{mL}).

4) Turbine cooling, bleed flow, & leakage effects neglected.

5) No power is removed from the turbine to drive accessories (η_{mH} & η_{mL} are const.)

6) γ_t, C_{p_t} do not vary with the power setting (T_{t4}).

7) $1+f$ is considered as a constant.

8.1.2 Dimensionless & corrected Performance parameters

Dimensional Analysis identifies correlating parameters that allow data taken under one set of cond^{n}s to be extended to other conditions.

Dimensionless pressure ⟶ δ

Dimensionless temp. ⟶ θ

$$\delta_i = \frac{P_{t_i}}{P_{ref}} \quad \text{(stagnation pressure at station (i))} \qquad (8.1a)$$

$$\theta_i = \frac{T_{t_i}}{T_{ref}} \qquad (8.1b)$$

Corrected mass flow rate (i): $\dot{m}_{c_i} = \dfrac{\dot{m}_i \sqrt{\theta_i}}{\delta_i}$ (8.2)

$$\dot{m}_i = \rho_i A_i V_i = \frac{P_{ti} A_i}{\sqrt{T_{ti}}} \times MFP(M_i)$$

$$\frac{\dot{m}_{ci}}{A_i} = \frac{\dot{m}_i \sqrt{T_{ti}}}{A_i P_{ti}} \frac{P_{std}}{\sqrt{T_{std}}} = \frac{P_{std}}{\sqrt{T_{std}}} MFP(M_i) \qquad \text{——— (8.3)}$$

$$MFP(M_i) = \sqrt{\frac{\gamma}{R}} M \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

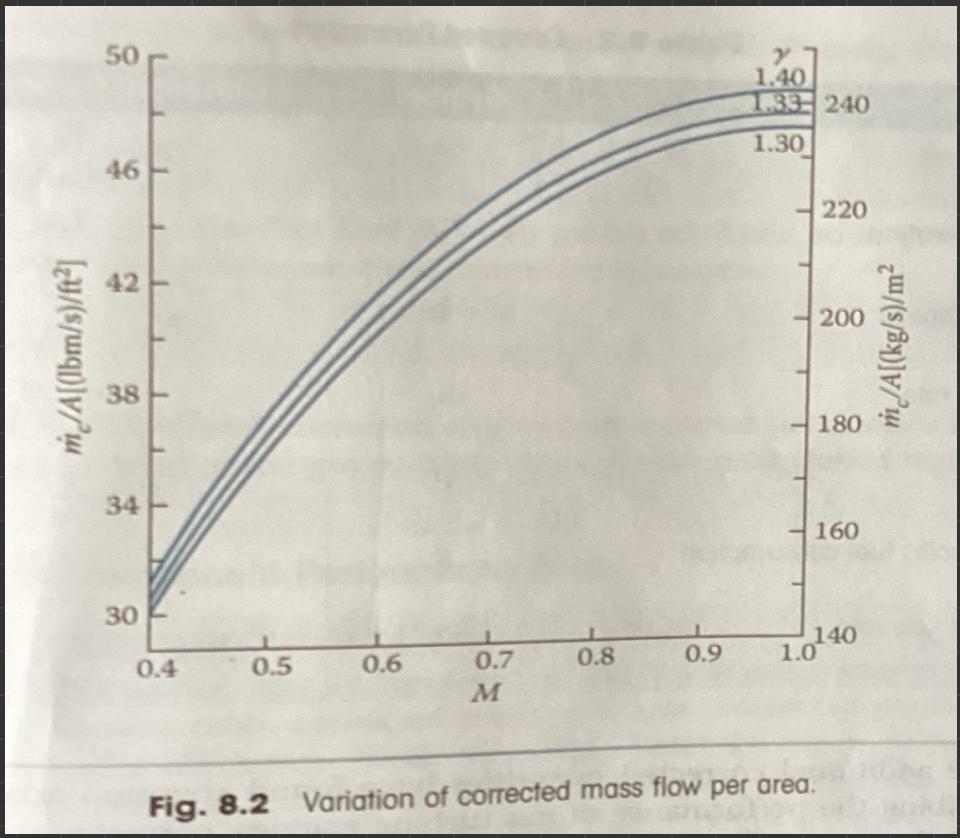


Fig. 8.2 Variation of corrected mass flow per area.

At station 2, M_2 is generally limited to 0.65 (to prevent excessive losses)

$M_2 = 0.65$, ⟶ $\dot{m}_{c_2}/A_2 = 214.8$ kg/s.m² {From graph}

Corrected engine speed (i): $N_{c_i} = \frac{N}{\sqrt{\theta_i}}$ ← RPM

Table 8.2 Corrected Parameters

Parameter	Symbol	Corrected Parameter
Total pressure	P_{ti}	$\delta_i = \frac{P_{ti}}{P_{\text{ref}}}$
Total temperature	T_{ti}	$\theta_i = \frac{T_{ti}}{T_{\text{ref}}}$
Rotational speed	$N = \text{RPM}$	$N_{ci} = \frac{N}{\sqrt{\theta_i}}$
Mass flow rate	$\dot{m}_i$	$\dot{m}_{ci} = \frac{\dot{m}_i \sqrt{\theta_i}}{\delta_i}$
Thrust	F	$F_c = \frac{F}{\delta_0}$
Thrust specific fuel consumption	S	$S_c = \frac{S}{\sqrt{\theta_0}}$
Fuel mass flow rate	$\dot{m}_f$	$\dot{m}_{fc} = \frac{\dot{m}_f}{\delta_2 \sqrt{\theta_2}}$

The corrected quantities collapse the variation in fuel consumption with flight cond^n & throttle setting into a single curve or set of curves.

$$\dot{m}_{f_c} = \frac{\dot{m}_f}{\delta_2 \sqrt{\theta_2}} \quad ; \quad S = \frac{\dot{m}_f}{F} \quad \Rightarrow \quad S_c = \Pi_d \frac{\dot{m}_{f_c}}{F_c}$$

8.1.3 Component Performance Maps

i) Compressor & Fan Performance maps:

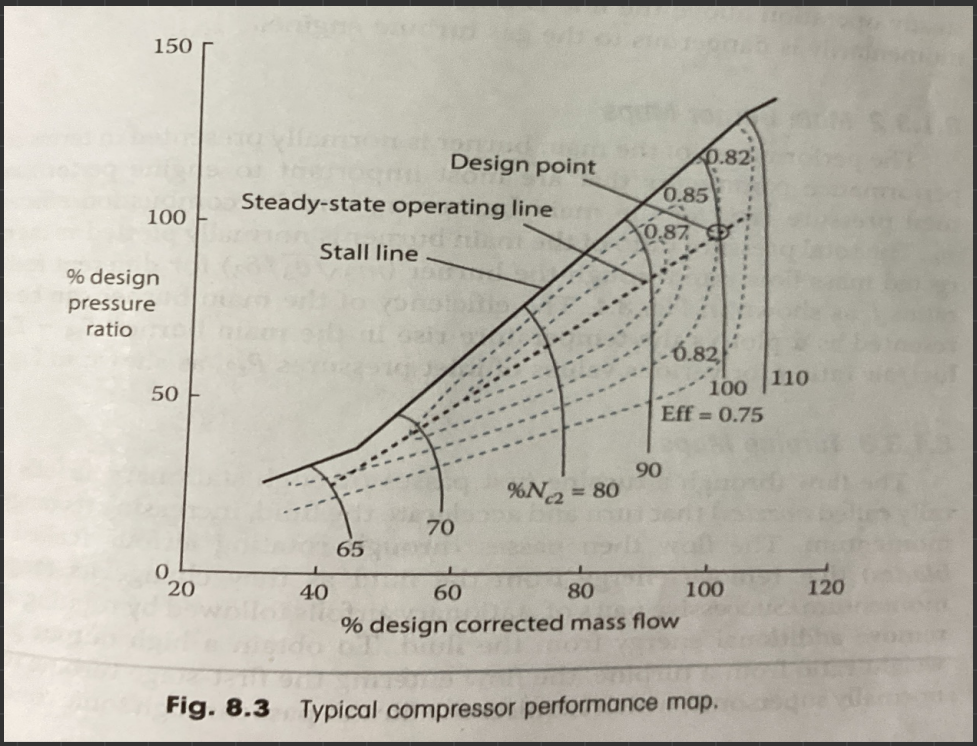


Fig. 8.3 Typical compressor performance map.

A limitation on fan & compressor performance of special concern is the stall or surge line.
Basically, in the map, we cannot go above the stall line.

ii) Main Burner Maps:

f ↑, η_b initially ↑
But we have T_{t4} limit, ∴ if f ↑ too much, η_b goes down after a pt.

P_{t3} ↑, T_{t3} ↑
somehow, η_b ↑.

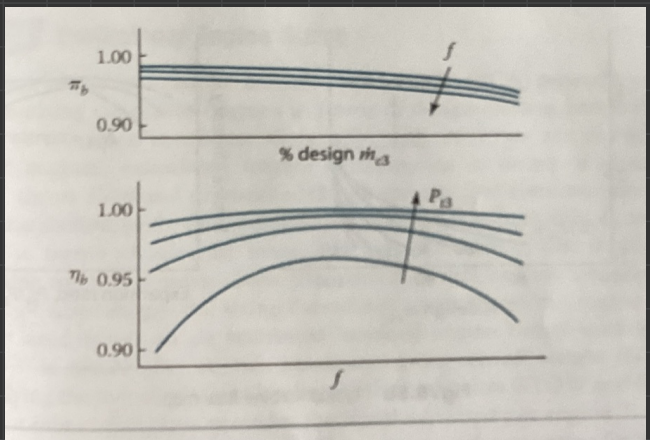


Fig. 8.4 Combustor performance.

iii) Turbine maps :

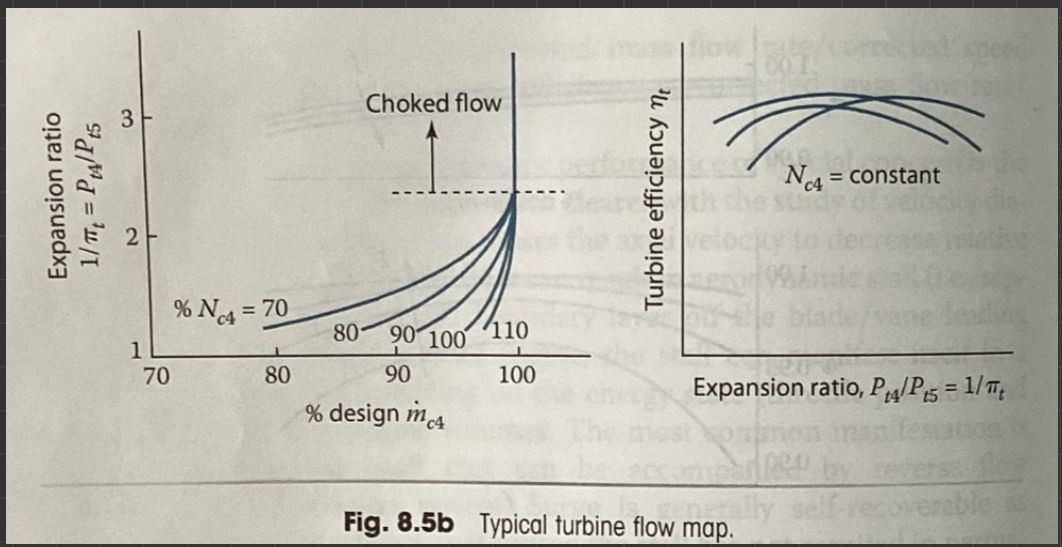


Fig. 8.5b Typical turbine flow map.

$\dot{m}_{c_4}$ ↑, for fixed N_{c_4}, $1/\pi_t$ ↑.

As more mass flows through the turbine, more work is extracted?

8.2 Gas Generator or Core

8.2.1 Conservⁿ of mass: Single spool Turbojet :

A_4 = Const.

$$\dot{m}_2 + \dot{m}_f = (1+f)\dot{m}_2 = \dot{m}_4 = \frac{P_{t4} A_4}{\sqrt{T_{t4}}} MFP(M_4)$$

Also, $$\dot{m}_{c_2} = \dot{m}_2 \frac{\sqrt{T_{t2}}}{P_{t2}} \frac{P_{ref}}{\sqrt{T_{ref}}}$$

$$\Rightarrow \dot{m}_{c_2} = \sqrt{\frac{T_{t2}}{T_{ref}}} \frac{P_{ref}}{P_{t2}} \frac{P_{t4}}{\sqrt{T_{t4}}} \frac{A_4}{1+f} MFP(M_4)$$

$\frac{P_{t4}}{P_{t2}} = \pi_c \pi_b$

$$\therefore \dot{m}_{c_2} = \sqrt{\frac{T_{t2}}{T_{t4}}} \pi_c \pi_b \frac{P_{ref}}{\sqrt{T_{ref}}} \frac{A_4}{1+f} MFP(M_4) \quad \text{——— (8.9)}$$

For const. values of $\frac{T_{t4}}{T_{t2}}$, A_4, f, & M_4, we have the following graph.

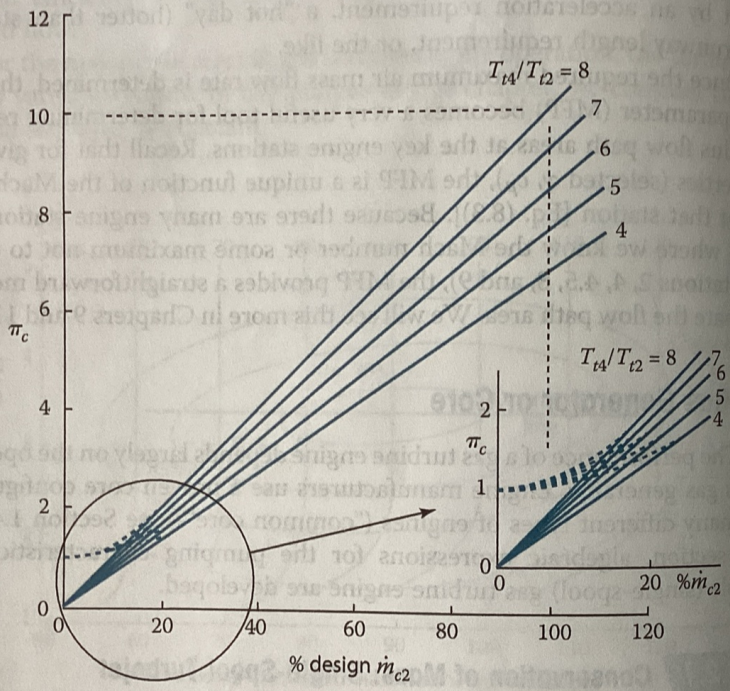


Fig. 8.6 Compressor map with lines of constant T_{t4}/T_{t2}.

These lines start at a pressure ratio of 1 and corrected mass flow rate of 0 & are curved for low compressor pressure ratios because station 4 is unchoked.

Rewriting eq. (8.9), $\dot{m}_{c_2} = C_1 \dfrac{\pi_c}{\sqrt{T_{t4}/T_{t2}}}$

where, $C_1 = \dfrac{\pi_b P_{ref}}{\sqrt{T_{ref}}} \dfrac{A_4}{1+f} MFP(M_4 = 1)$

For a gas generator, the specific relationship b/w the compressor pressure ratio and corrected mass flow rate is called the engine operating line & depends on the characteristics of the turbine.

8.2.2 Turbine Characteristics

$$\frac{\dot{m}_i\sqrt{T_{ti}}}{P_{ti}A_i} = MFP(M_i) = \sqrt{\frac{\gamma_i}{R_i}}\,M_i\left(1+\frac{\gamma_i-1}{2}M_i^2\right)^{\frac{-\gamma_i+1}{2(\gamma_i-1)}}$$

For $M_i = 1$,

$$\frac{\dot{m}_i\sqrt{T_{ti}}}{P_{ti}A_i} = MFP(M_i=1) = \sqrt{\frac{\gamma_i}{R_i}}\left(\frac{2}{\gamma_i+1}\right)^{\frac{\gamma_i+1}{2(\gamma_i-1)}} = \frac{\Gamma_i}{\sqrt{R_i}}$$

$$\Gamma_i = \sqrt{\gamma_i}\left(\frac{2}{\gamma_i+1}\right)^{\frac{\gamma_i+1}{2(\gamma_i-1)}}$$

For a turbojet engine, flow is choked at stations 4 & 8.

$$\dot{m}_4 = \frac{P_{t4}A_4}{\sqrt{T_{t4}}}\frac{\Gamma_4}{\sqrt{R_4}}, \quad \dot{m}_8 = \frac{P_{t8}A_8}{\sqrt{T_{t8}}}MFP(M_8)$$

$$\dot{m}_4 = \dot{m}_8 \Rightarrow \frac{\sqrt{\tau_t}}{\pi_t} = \frac{A_8}{A_4}\frac{MFP(M_8)}{\Gamma_4/\sqrt{R_4}} \qquad (8.12a)$$

where,

$$\pi_t = \left(1-\frac{1-\tau_t}{\eta_t}\right)^{\gamma_t/\gamma_t-1} \qquad (8.12b)$$

(η_t: isentropic efficiency)

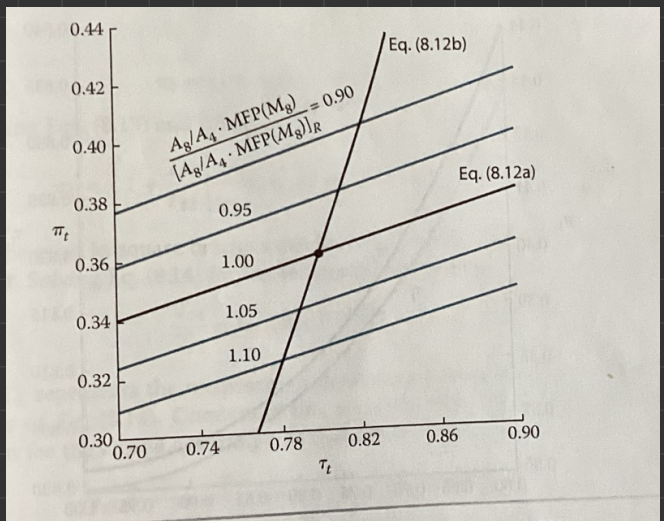
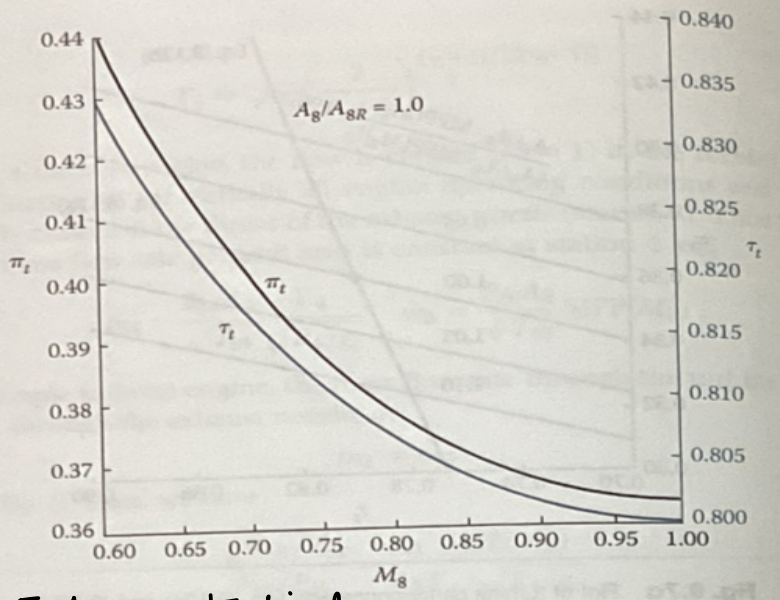


Fig. 8.7a Plot of turbine performance with Eqs. (8.12a) and (8.12b).



$M_8 \downarrow$, $\tau_t \uparrow \longrightarrow \downarrow$ Turbine Power

Fig. 8.7b Variation of turbine performance with exhaust nozzle Mach number.

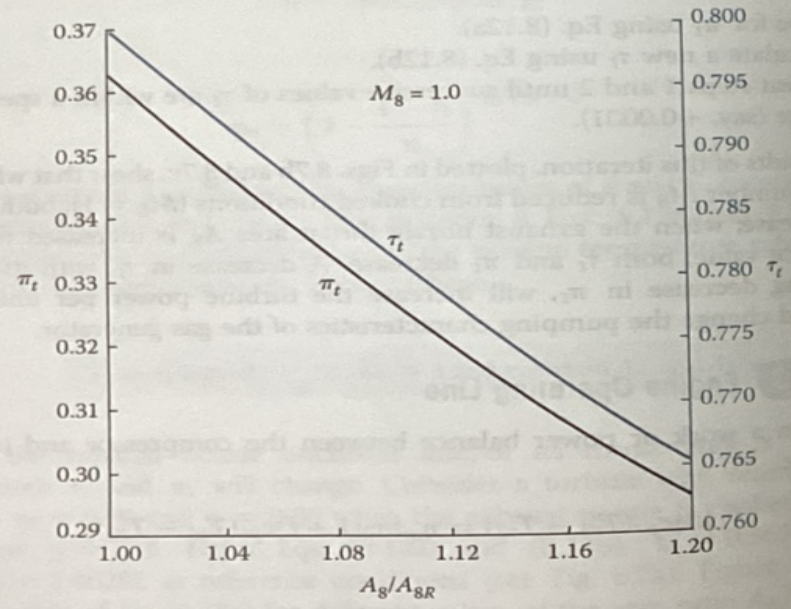


Fig. 8.7c Variation of turbine performance with exhaust nozzle area.

8.2.3 Engine Operating Line

From Energy balance b/w compressor & turbine,

$$\dot{m}_2 c_{pc}(T_{t3} - T_{t2}) = \eta_m \dot{m}_2 (1+f) c_{pt}(T_{t4} - T_{t5})$$

$$\tau_c = 1 + \frac{T_{t4}}{T_{t2}} \frac{c_{pt}}{c_{pc}} \eta_m (1+f)(1-\tau_t)$$

$$\pi_c = [1 + \eta_c(\tau_c - 1)]^{\gamma_c/\gamma_c - 1}$$

$$\pi_c = \left\{1 + \frac{T_{t4}}{T_{t2}} \underbrace{\left[\frac{c_{pt}}{c_{pc}} \eta_c \eta_m (1+f)(1-\tau_t)\right]}_{1/C_2}\right\}^{\frac{\gamma_c}{\gamma_c - 1}}$$

when τ_t is considered a const.,

$$\Rightarrow \frac{T_{t4}}{T_{t2}} = C_2\left[(\pi_c)^{\frac{\gamma_c - 1}{\gamma_c}} - 1\right]$$

$$\dot{m}_{c_2} = \frac{C_1}{\sqrt{C_2}} \frac{\pi_c}{\sqrt{\pi_c^{\gamma_c - 1/\gamma_c} - 1}}$$

} eqⁿ for engine operating line.

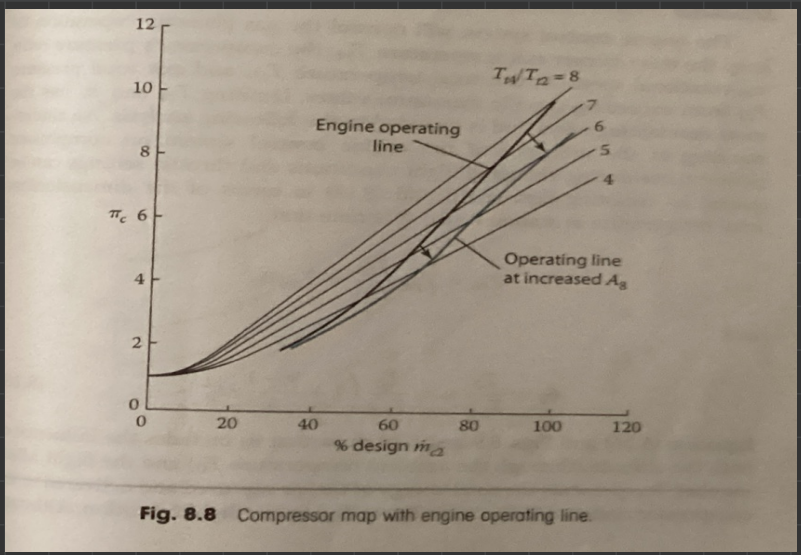


Fig. 8.8 Compressor map with engine operating line.

The engine operating line shows that for a fixed T_{t4}/T_{t2}, there is one value of π_c & $\dot{m}_{c_2}$.

T_{t_2} ⟶ fixed, : π_c, $\dot{m}_{c_2}$ ↑ with T_{t4} ↑
T_{t4} ⟶ fixed, : π_c, $\dot{m}_{c_2}$ ↓ with T_{t_2} ↑

Curving of line below $\pi_c = 4$ is due to $M_8 < 1$, which ↑ τ_t.

8.2.4 Engine Controls

The engine control system will control gas generator operation to keep T_{t4}, π_c, N, T_{t3}, & P_{t3} from exceeding specific max^m values.

limiting T_{t4} & π_c ⟶ most dominating effect.

$$T_{t0} = T_{ref}\,\theta_0$$

$$\theta_0 = \frac{T_0}{T_{ref}}\,\tau_r = \frac{T_0}{T_{ref}}\left(1 + \frac{\gamma-1}{2} M_0^2\right)$$

↳ θ_0 includes the influence of both, the altitude & flight mach no. (M_0).

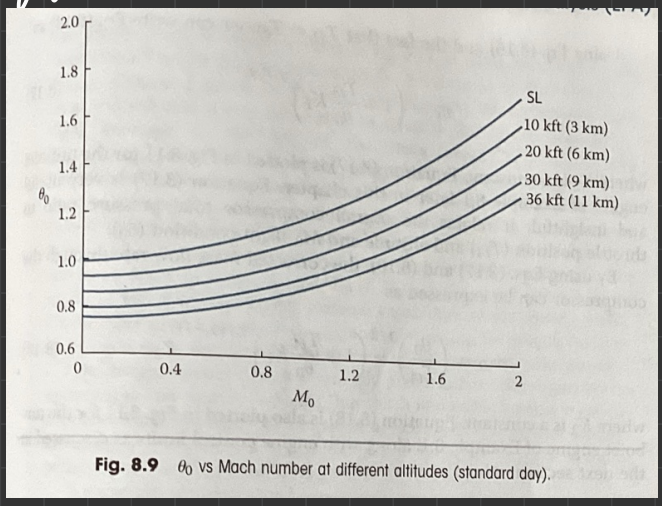


Fig. 8.9 θ_0 vs Mach number at different altitudes (standard day).

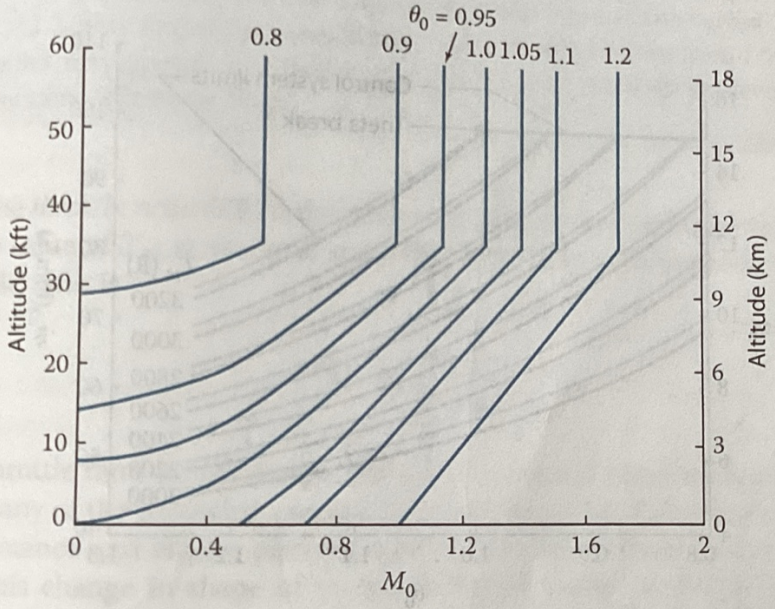


Fig. 8.10 θ_0 vs Mach number and altitude (standard day).

In the previous section, we discussed:

$$\pi_c = \left\{1 + \frac{T_{t4}}{T_{t2}}\left[\frac{c_{pt}}{c_{pc}}\eta_c\eta_m(1+f)(1-\tau_t)\right]\right\}^{\frac{\gamma_c}{\gamma_c-1}}$$

Using $T_{t2} = T_{t0}$,

$$\pi_c = \left(1 + \frac{T_{t4}}{\theta_0}K_1\right)^{\frac{\gamma_c}{\gamma_c-1}} \qquad \text{(eq. 8.17)}$$

(K_1: const.)

This eqⁿ relates π_c to throttle posⁿ, altitude, & M_0.

Also, $\dot{m}_{c_2} = C_1 \dfrac{\pi_c}{\sqrt{T_{t4}/T_{t2}}}$

$$\Rightarrow \dot{m}_{c_2} = \sqrt{\frac{\theta_0}{T_{t4}}}\left(1 + \frac{T_{t4}}{\theta_0}K_1\right)^{\frac{\gamma_c}{\gamma_c-1}} K_2$$

(K_2: const.)

8.2.5 Theta Break & Throttle ratio

The throttle ratio (TR) = $\frac{(T_{t4})_{max}}{(T_{t4})_{SLS}}$

eq. 8.17 is plotted here

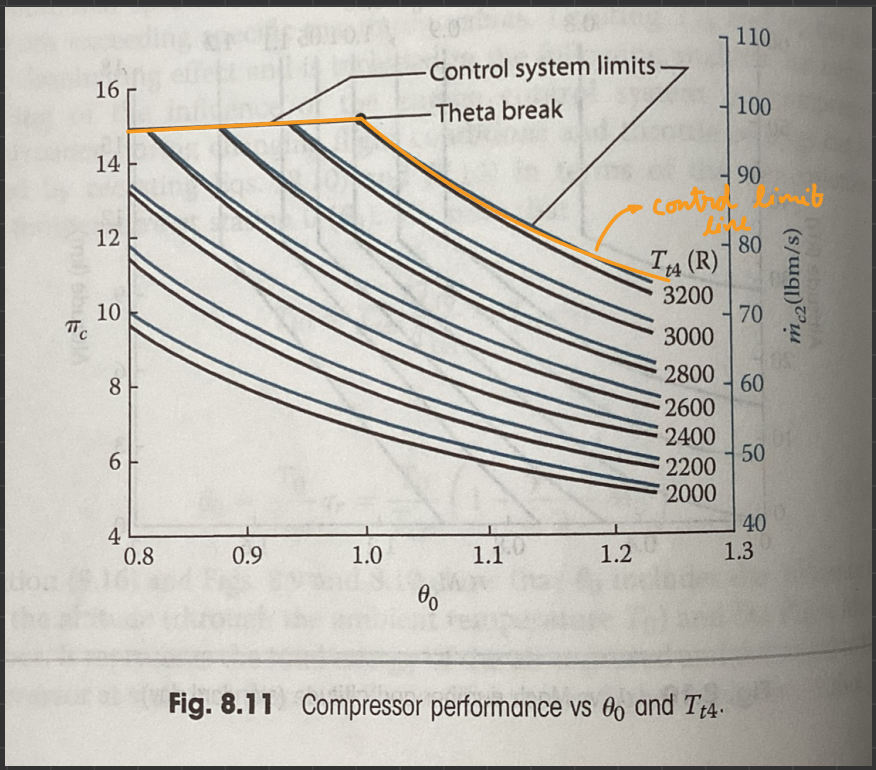


Fig. 8.11 Compressor performance vs θ_0 and T_{t4}.

At any pt. on the control limit line to the left of the θ_0 break

$\pi_c = \pi_{c\,max}$, & $T_{t4} < T_{t4\,max}$

↳ Engine operates on π_c limit

At any pt. on the control limit line to the right of θ_0 break,

$T_{t4} = T_{t4\,max}$, & $\pi_c < \pi_{c\,max}$

↳ Engine operates on T_{t4} limit.

$\theta_0 < \theta_{0\,break}$ & $T_{t4} < T_{t4\,max}$ ⟶ ST ↓ than inherent mat. capab.

$\theta_0 > \theta_{0\,break}$ & $\pi_c < \pi_{c\,max}$ ⟶ TSFC ↑ than η_{th} makes possible.

$TR = \theta_0$ at max π_c & max T_{t4}

$\Rightarrow TR = \theta_{0\,break}$

8.2.6 Variation in Engine speed

For a CPG,

$$T_{t3} - T_{t2} = K_1 N^2$$ ← rotational speed

$$\tau_c - 1 = \frac{K_1 N_{c_2}^2}{T_{ref}} \quad \text{——— i)}$$

$$\tau_c - 1 = \left(\pi_c^{(\gamma_c-1)/\gamma_c} - 1\right)/\eta_c$$

$$\frac{\pi_c^{(\gamma_c-1)/\gamma_c} - 1}{N_{c_2}^2} = \frac{\eta_c K_1}{T_{ref}} = \frac{\pi_{cR}^{(\gamma_c-1)/\gamma_c} - 1}{N_{c_2R}^2}$$

($\frac{\eta_c K_1}{T_{ref}}$: Const.; π_{cR}: Reference eng.)

$$\frac{N_{c_2}}{N_{c_2R}} = \sqrt{\frac{\pi_c^{(\gamma_c-1)/\gamma_c} - 1}{\pi_{cR}^{(\gamma_c-1)/\gamma_c} - 1}} \quad \text{——— (8.20a)}$$

Using eq. (8.17), $\frac{N_{c_2}}{N_{c_2R}} = \sqrt{\frac{(T_{t4}/\theta_0)}{(T_{t4}/\theta_0)_R}}$ ——— (8.20b)

→ engine throttle & flight cond^n.

Removing the "Corrected parameters",

$$\frac{N}{N_R} = \sqrt{\frac{T_{t4}}{T_{t4R}}} \quad \text{——— (8.21a)}$$

$$\text{Now, } \frac{N_{max}}{N_{SLS}} = \sqrt{\frac{T_{t4\,max}}{T_{t4\,SLS}}} = \sqrt{TR} = \sqrt{\theta_{0\,break}} \quad \text{——— (8.21b)}$$

→ overspeed

$TR = 1.08 \rightarrow$ overspeed ≈ 1.04

$\therefore N_{max} \rightarrow 4\%$ faster than N_{SLS}

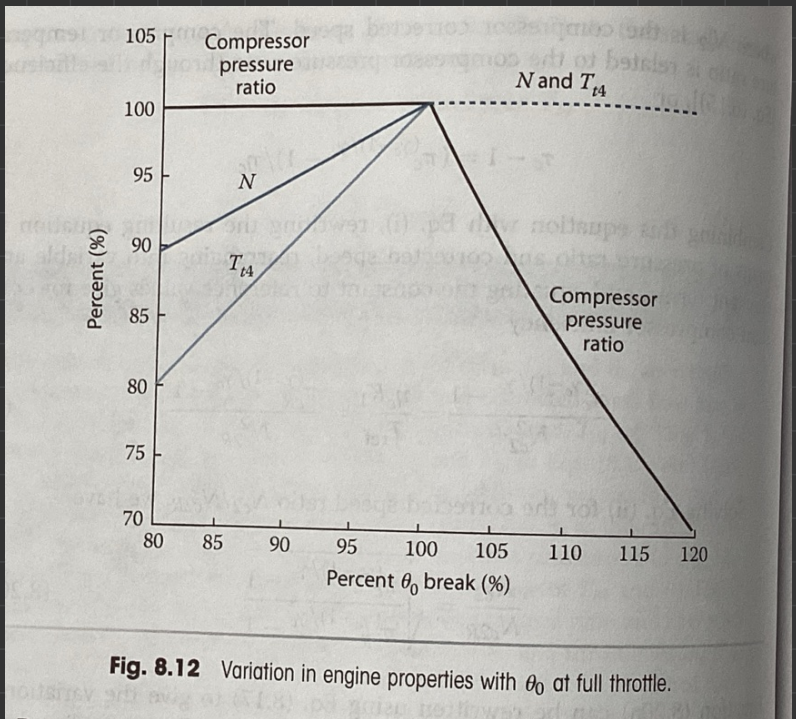


Fig. 8.12 Variation in engine properties with θ_0 at full throttle.

Compressor & turbine have same shaft, ∴ same N.

$$N_{C_2} = \frac{1}{\sqrt{T_{ref}}} \sqrt{\frac{T_{t4}}{\theta_0}}\, N_{C_4}$$

From (8.20b),

N_{C_4} = const. } → τ_b, π_t, η_t = const.

8.2.7 Gas Generator eq^n's

pumping characteristics of a simple gas generator can be represented by the variation of the gas gen.'s parameter ratios with corrected compressor speed.

$$\frac{P_{t6}}{P_{t_2}} = \pi_c \pi_b \pi_t \quad \text{——— (8.23)}$$

$$\frac{T_{t6}}{T_{t_2}} = \frac{T_{t4}}{T_{t2}} \tau_t \quad \text{——— (8.24)}$$

$$\frac{\dot{m}_{C_2}}{\dot{m}_{C_{2R}}} = \frac{\pi_c}{\pi_{cR}} \sqrt{\frac{(T_{t4}/T_{t2})_R}{T_{t4}/T_{t_2}}} \quad \text{——— (8.25)}$$

From eq. (8.17),

$$\pi_c = \left[1 + \frac{T_{t4}/T_{t2}}{(T_{t4}/T_{t2})_R}\left(\pi_{cR}^{(\gamma_c-1)/\gamma_c} - 1\right)\right]^{\gamma_c/(\gamma_c-1)} \qquad (8.26)$$

$$\frac{N_{c2}}{N_{c2R}} = \sqrt{\frac{T_{t4}/T_{t2}}{(T_{t4}/T_{t2})_R}} \qquad (8.27)$$

$$\dot{m}_f = \dot{m}_0 \frac{h_{t4} - h_{t3}}{\eta_b h_{pR} - h_{t4}} \quad ; \quad \dot{m}_{fc} = \frac{\dot{m}_{c2}}{\theta_2} \frac{h_{t4} - h_{t3}}{\eta_b h_{pR} - h_{t4}} \qquad (8.28)$$

$$\tau_c = 1 + (\tau_{cR} - 1)\frac{T_{t4}/T_{t2}}{(T_{t4}/T_{t2})_R} \qquad (8.29)$$

The above eq^{n}s are functions of T_{t4}/T_{t2}, except for (8.28).

8.3 Turbojet Engine

We assume choked flow at stations 4 & 8.
$M_0, T_0, P_0, T_{t4}, P_9/P_9$ can be independently varied.

Table 8.3 Performance Analysis Variables for Single-Spool Turbojet Engine

Component	Variables		
	Independent	Constant or Known	Dependent
Engine	M_0, T_0, P_0		$\dot{m}_0$
Diffuser		$\pi_d = f(M_0)$	
Compressor		η_c	π_c, τ_c
Burner	T_{t4}	π_b, η_b	f
Turbine		π_t, τ_t	
Nozzle	P_9/P_0	π_n	$M_9, T_9/T_0$
Total number	5		6

From PCA, we get the following 7 eq^{n}s for a Turbojet.

i) $$\frac{F}{\dot{m}_0} = a_0\left[(1+f)\frac{V_9}{a_0} - M_0 + (1+f)\frac{R_t}{R_c}\frac{T_9/T_0}{V_9/a_0}\left\{1 - \frac{P_0/P_9}{r_c}\right\}\right]$$ {Specific Thrust}

ii) $$\frac{T_9}{T_0} = \frac{T_{t4}\tau_t}{(P_{t9}/P_9)^{(r_t-1)/r_c}}$$

iii) $$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9}\pi_r\pi_d\pi_c\pi_b\pi_t\pi_n$$

iv) $$M_9 = \sqrt{\frac{2}{r_t - 1}\left[\left(\frac{P_{t9}}{P_9}\right)^{(r_t-1)/r_t} - 1\right]}$$

v) $$\frac{V_9}{a_0} = M_9\sqrt{\frac{r_t R_t T_9}{r_c R_c T_0}}$$

vi) $$TSFC = f/F/\dot{m}_0$$

vii) $$f = \frac{h_{t4} - h_{t3}}{h_{PR}\eta_b - h_{t4}}$$

All the above 7 eq^{n}s can be solved for given $T_{t4}, M_0, T_0, P_0, P_0/P_9$ with expressions for $\tau_\lambda, \pi_r, \tau_r, \pi_d, \pi_c, \tau_c$.

eq. 8.29 →

$$\tau_c = 1 + (\tau_{cR} - 1)\frac{T_{t4}/T_{t2}}{(T_{t4}/T_{t2})_R}$$

Also, $\dot{m}_0 = \frac{P_{t4}}{\sqrt{T_{t4}}}\frac{A_4}{1+f}\,MFP(1) = \frac{P_0 \pi_r \pi_d \pi_c}{\sqrt{T_{t4}}}\underbrace{\left[\frac{\pi_b A_4}{1+f}\,MFP(1)\right]}_{\text{Const.}}$

$$\Rightarrow \quad \frac{\dot{m}_0\sqrt{T_{t4}}}{P_0 \pi_r \pi_d \pi_c} = \frac{\pi_b A_4}{1+f}\,MFP(1) = \left(\frac{\dot{m}_0\sqrt{T_{t4}}}{P_0 \pi_r \pi_d \pi_c}\right)_R$$

$$\dot{m}_0 = \dot{m}_{0R}\,\frac{P_0 \pi_r \pi_d \pi_c}{(P_0 \pi_r \pi_d \pi_c)_R}\sqrt{\frac{T_{t4R}}{T_{t4}}}$$

Throat area of exhaust nozzle A_8 is assumed to be const.

P_0/P_9 — independent variable

A_9 must correspond to the nozzle pressure ratio P_{t9}/P_9

from eq. (8.11), {A_9 changes acc. to P_{t9}/P_9}

i) —— $\dot{m}_8 = \frac{P_{t8} A_8}{\sqrt{T_{t8}}}\frac{\Gamma_t}{\sqrt{R_t}}$ {station 8 is choked}

ii) —— $\dot{m}_9 = \frac{P_{t9} A_9}{\sqrt{T_{t9}}}\sqrt{\frac{\gamma_t}{R_t}}\,M_9\left(1 + \frac{\gamma_t - 1}{2}M_9^2\right)^{-\frac{\gamma_t+1}{2(\gamma_t-1)}}$

Nozzle relationship: $T_{t8} = T_{t9}$, $\pi_n = P_{t9}/P_{t7}$; $\dot{m}_8 = \dot{m}_9$

$$\Rightarrow \quad \frac{A_9}{A_8} = \frac{\Gamma_t}{\sqrt{\gamma_t}}\frac{1}{\pi_n}\frac{1}{M_9}\left(1 + \frac{\gamma_t - 1}{2}M_9^2\right)^{(\gamma_t+1)/[2(\gamma_t-1)]}$$

$$M_9 = \sqrt{\frac{2}{\gamma_t - 1}\left[\left(P_{t9}/P_9\right)^{\gamma_t - 1/\gamma_t} - 1\right]}$$

$$\Rightarrow \frac{A_9}{A_8} = \Gamma_t \sqrt{\frac{\gamma_t - 1}{2\gamma_t}} \frac{1}{\pi_n} \frac{\left(P_{t9}/P_9\right)^{\frac{(\gamma_t+1)}{2\gamma_t}}}{\sqrt{\left(P_{t9}/P_9\right)^{(\gamma_t-1)/\gamma_t} - 1}} \quad \text{——— (8.31)}$$

Again, since A_8 is const, we can find A_9 from A_{9R}

$$\frac{A_9}{A_{9R}} = \left[\frac{P_{t9}/P_9}{(P_{t9}/P_9)_R}\right]^{(\gamma_t+1)/2\gamma_t} \sqrt{\frac{(P_{t9}/P_9)_R^{(\gamma_t-1)/\gamma_t} - 1}{(P_{t9}/P_9)^{(\gamma_t-1)/\gamma_t} - 1}} \quad \text{——— (8.32)}$$

8.3.1 Corrected Engine Performance & value of Throttle ratio

$$F = \dot{m}_0\left[(1+f)V_9 - V_0\right]$$

where, $V_9 = M_9 \times a_9 = \sqrt{2 C_{pt} T_{t4} \tau_t \left[1 - \left(\pi_r \pi_d \pi_c \pi_b \pi_t \pi_n\right)^{-(\gamma_t-1)/\gamma_t}\right]}$

$$V_0 = M_0 a_0 = M_0 \sqrt{\gamma_c R_c T_0}$$

$$\dot{m}_0 = \dot{m}_{c0} \frac{\delta_0}{\sqrt{\theta_0}} = \dot{m}_{c_2} \frac{\delta_2}{\sqrt{\theta_2}} = \dot{m}_{c_2} \frac{\pi_d \delta_0}{\sqrt{\theta_0}}$$

Compressor corrected mass flow rate

$$\Rightarrow \quad F = \dot{m}_{c_2} \frac{\pi_d \delta_0}{\sqrt{\theta_0}}\left[(1+f)V_9 - V_0\right]$$

$$\frac{F}{\delta_0} = \dot{m}_{c_2} \pi_d \left[(1+f)\frac{V_9}{\sqrt{\theta_0}} - \frac{V_0}{\sqrt{\theta_0}}\right] \quad \text{——— (8.33)}$$

where,

$$\frac{V_9}{\sqrt{\theta_0}} = \sqrt{\frac{T_{t4}}{T_{t2}}} \sqrt{2 C_{pt} T_{ref} \tau_t \left[1 - \left(\pi_r \pi_d \pi_c \pi_b \pi_t \pi_n\right)^{-(\gamma_t-1)/\gamma_t}\right]}$$

&

$$\frac{V_0}{\sqrt{\theta_0}} = \frac{M_0}{\sqrt{\tau_r}} a_{SL}$$ → speed of sound at SLS cond^{n}s.

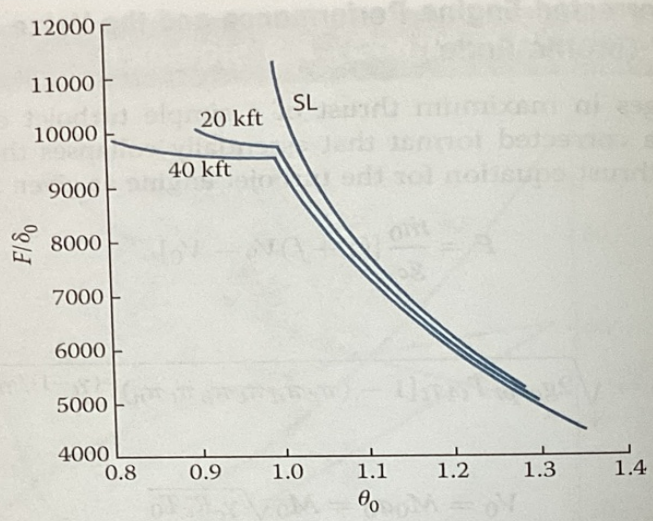


Fig. 8.19 Maximum corrected thrust (F/δ_0) of a turbojet vs θ_0.

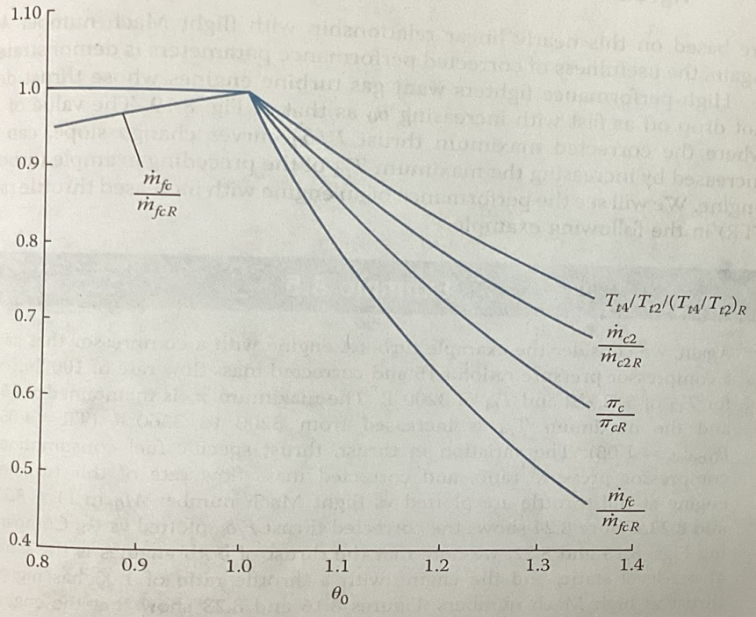


Fig. 8.20 Maximum throttle characteristics of a turbojet vs θ_0.

Because the compressor & turbine are connected to the same shaft, they have the same rotational speed N.

$$N_{C_2} = \frac{1}{N_{C_4}\sqrt{\frac{T_{t4}}{\theta_0}\frac{T_{ref}}{}}}$$

8.3.2.1 High Pressure Turbine

$$\dot{m}_4 = \dot{m}_{4.5}$$

$$\dot{m}_4 = \frac{P_{t4}}{\sqrt{T_{t4}}} A_4 \, MFP(M_4) = \dot{m}_{4.5} = \frac{P_{t4.5}}{\sqrt{T_{t4.5}}} A_{4.5} \, MFP(M_{4.5})$$

Assumptions: choked flow at stations 4 & 4.5.
Areas are const.

$$\frac{P_{t4}/P_{t4.5}}{\sqrt{T_{t4}/T_{t4.5}}} = \frac{\sqrt{\tau_{tH}}}{\pi_{tH}} = \text{const.}$$

$\eta_{tH} = \text{const.} \Rightarrow \pi_{tH}, \tau_{tH}, \dot{m}_{C_4}$, & $\dot{m}_{C_{4.5}}$ are const.

8.3.2.2 Low Pressure Turbine

$$\dot{m}_{4.5} = \dot{m}_8$$

same assumptions as above.

$$\frac{P_{t4.5}/P_{t5}}{\sqrt{T_{t4.5}/T_{t8}}} \frac{1}{MFP(M_8)} = \frac{\sqrt{\tau_{tL}}/\pi_{tL}}{MFP(M_8)} = \frac{A_8 \pi_{AB}}{A_{4.5} MFP(M_{4.5})}$$

We can equate the const. term on the right to reference cond^{n}s.

$$\pi_{tL} = \pi_{tLR} \sqrt{\frac{\tau_{tL}}{\tau_{tLR}}} \frac{MFP(M_{8R})}{MFP(M_8)} \quad \text{——— (8.37)}$$

$$\tau_{tL} = 1 - \eta_{tL}\left(1 - \pi_{tL}^{(\gamma_t - 1)/\gamma_t}\right) \quad \text{——— (8.38)}$$